

MODULE 47 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 3

Fifteen questions on improving what already exists.

DOMAIN 3 of 4 EXAM WEIGHT 25% SET 3 of 5

A connector on-premises publishes a custom heartbeat metric to CloudWatch every minute while it is healthy. An alarm is configured to fire if the metric drops below one. The connector's host lost power for nine hours overnight and the alarm never fired. The threshold and evaluation period were both reviewed and are correct.

What should the solutions architect change?

- ☐ A Increase the number of evaluation periods
- ☐ B Publish the metric at high resolution
- ☐ C Set the alarm's missing data treatment to breaching
- ☐ D Add a second alarm on the same metric with a lower threshold

◆ PAUSE AND DECIDE

What should the solutions architect change?

- A

 More evaluation periods
Evaluating over longer still evaluates nothing. The absence of data points is the problem, not their number.
- B

 High resolution
Publishes more often while healthy. When the host is off, there is nothing to publish at any resolution.
- C

 Missing data treated as breaching
With no host there is no data point, so the alarm sits in INSUFFICIENT_DATA and never pages. Breaching makes absence the alarm condition.
- D

 Second alarm, lower threshold
A second alarm on the same absent metric behaves identically. Two silences instead of one.

THE SIGNAL

A metric that exists only while something is healthy disappears exactly when you need it. Default treatment is the trap. Module 23.

An Auto Scaling group must scale on memory utilisation, because the workload is memory-bound and CPU stays below 25% throughout. The team enabled EC2 detailed monitoring and still cannot find a memory metric to build a scaling policy on.

The instances run Amazon Linux 2023.

What is required?

- ☐ A Install and configure the CloudWatch agent to publish memory as a custom metric
- ☐ B Enable Compute Optimizer to expose memory metrics
- ☐ C Switch the scaling policy to a target tracking policy on CPU
- ☐ D Enable Container Insights on the Auto Scaling group

◆ PAUSE AND DECIDE

What is required?

- A

 CloudWatch agent
Memory lives inside the guest OS, which the hypervisor cannot observe. Only an agent on the instance can publish it.
- B

 Compute Optimizer
Makes rightsizing recommendations and can consume memory data. It does not generate the metric, and it is not a scaling source.
- C

 Scale on CPU
The scenario says CPU stays below 25% while memory is the constraint. Scaling on CPU would never trigger.
- D

 Container Insights
For ECS and EKS container metrics. Nothing suggests containers are involved.

THE SIGNAL

Detailed monitoring buys frequency, not coverage. It changes five minutes to one and adds no new metric. Module 23.

A checkout API has a p99 latency of six seconds against a one-second target. The request passes through eight microservices and two third-party calls. Every owning team reports that its own service's average latency is well within target, and three incident calls have ended without a cause.

The services are instrumented with nothing beyond CloudWatch metrics.

What should the solutions architect implement?

- ☐ A CloudWatch dashboard combining all eight services' latency metrics
- ☐ Distributed tracing with AWS X-Ray, instrumented through ADOT, and review the service map
- ☐ Detailed monitoring on every service's instances
- ☐ Increased log verbosity across all eight services

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Combined dashboard

The strong near miss. Eight averages on one screen still cannot connect the slow requests to one another.

C Detailed monitoring

Higher resolution on the same per-service averages, which every team already reports as healthy.

B X-Ray tracing via ADOT

A trace follows one slow request through every hop, including the third parties, and attributes the milliseconds. Only tracing answers where.

D More logging

Eight log streams with no shared request identifier. Correlating one journey stays manual.

THE SIGNAL

Healthy averages plus a bad p99 is not a contradiction. Each service is slow for a small, different subset of requests. Module 24.

A retailer's checkout failed for two hours at 03:00. Every internal health check passed: instances healthy, target groups healthy, database responsive. The cause was an expired certificate on a third-party payment script loaded by the browser. Customers reported it at 08:00.

The company already runs CloudWatch RUM.

What would have detected this?

- ☐ A More aggressive ALB health check thresholds
- ☐ B CloudWatch RUM, which is already deployed
- ☐ C X-Ray tracing on the checkout service
- ☐ D A CloudWatch Synthetics canary completing a checkout every five minutes

◆ PAUSE AND DECIDE

What would have detected this?

A Tighter health checks

Health checks probe your targets, which were genuinely healthy. No threshold detects a broken third-party asset.

C X-Ray

Traces server-side request paths. A certificate failing in the browser never reaches your services.

B RUM, already deployed

The planted trap. RUM would catch this class of fault, and it needs real users. At 03:00 there were none, which is why it did not.

D A checkout-flow canary

Runs on a schedule regardless of traffic, exercises the full journey from outside, and would have failed within five minutes.

THE SIGNAL

"RUM is already deployed" is planted to tempt you. 03:00 is the word that rules it out: real user monitoring needs real users. Module 24.

An audit finds 512 resources that an AWS Config remediation rule should have corrected over the past four months. The remediation runbook has been failing since an IAM permission change, and no alert was raised. The Config console showed the rule as configured with automatic remediation throughout.

Weekly compliance reports were being reviewed.

What should the solutions architect implement?

- ☐ A An EventBridge rule matching SSM Automation executions with status Failed or TimedOut, routed to an SNS topic
- ☐ B An increased remediation retry count and retry interval
- ☐ C Daily rather than weekly compliance report reviews
- ☐ D AdministratorAccess on the remediation role to prevent permission gaps

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

 Alarm on failed automations
Closes the fourth step of the remediation loop. A failing runbook becomes visible immediately rather than at the next audit.
- B

 More retries
Retrying a call the role has no permission to make fails identically, more often. The failure is deterministic.
- C

 Daily reviews
The console showed the rule as configured throughout, which is why nobody noticed. Reading a configuration view more often changes nothing.
- D

 AdministratorAccess
Closes this gap by removing least privilege from something that modifies production resources.

THE SIGNAL

Detect, route, act, verify. The fourth step is the one teams skip, and it fails silently for months. Module 25.

A bank uses Session Manager successfully. A new policy requires zero standing privileges to production nodes: every connection must be explicitly approved and time-bound at the moment of access. Non-production access must remain frictionless, and operators must not wait for approval there.

The estate includes on-premises managed nodes.

What should the solutions architect implement?

- ☐ A A permissions boundary limiting operators to non-production instances
- ☐ B Remove ssm:StartSession from all policies and grant it manually on request
- ☐ C Just-in-time node access, with approval policies requiring human approval for production and auto-approval elsewhere
- ☐ D A shorter Session Manager idle timeout on production

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Permissions boundary

Removes production access entirely rather than gating it. Operators legitimately need it sometimes.

B Manual IAM grants

Reaches the right outcome by the worst route: a human editing IAM at 3am, with no timeout, policy or audit structure.

C JITNA with per-environment policies

Approval policies can deny, auto-approve or require a human, so production is gated and non-production stays frictionless. Covers on-premises nodes too.

D Shorter idle timeout

Ends idle sessions sooner. The standing permission to start a new one at any moment is untouched.

THE SIGNAL

Two requirements pulling against each other, resolved by one mechanism with different policies per environment. Module 26.

Newly launched instances in a private subnet do not appear as managed nodes. The AMI includes the SSM Agent and the process is running. The instance profile has AmazonSSMManagedInstanceCore attached. There is no NAT gateway in the VPC. Identically configured instances in a public subnet register normally.

Security will not permit a NAT gateway.

What is required?

- ☐ A Attach AdministratorAccess to the instance profile
- ☐ B Create VPC interface endpoints for ssm, ssmmessages and ec2messages
- ☐ C Update the SSM Agent to the latest version
- ☐ D Enable Default Host Management Configuration for the account

◆ PAUSE AND DECIDE

What is required?

A AdministratorAccess

The managed instance core policy is exactly what AWS provides for this. More permission does not create a network path.

C Update the agent

The identical AMI registers in a public subnet, so the agent version is not the variable.

B Three interface endpoints

The agent connects outbound to Systems Manager. With no NAT and no endpoints, a private subnet has no route at all. Note there are three, not one.

D Default Host Management

An alternative way to grant permissions. The instances already have an instance profile.

THE SIGNAL

Same AMI, same role, different subnet: one variable. Missing ssmmessages is the classic partial fix that registers but cannot start a session. Module 26.

A platform team publishes a hardened golden AMI monthly using EC2 Image Builder. An audit finds that only 44% of running instances were launched from an approved image; the rest use images up to three years old. The company runs AWS Organizations with SCPs already in place.

Teams must still be able to launch instances.

What should the solutions architect recommend?

- ☐ A Email the current AMI ID to all teams monthly
- ☐ B Delete all non-approved AMIs from every account
- ☐ C Rely on Patch Manager to bring older instances up to date
- ☐ D An SCP denying `ec2:RunInstances` unless the AMI carries the approved tag, plus a Config rule detecting unapproved images already running

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Email the AMI ID

Depends on every team reading and acting on an email monthly. The audit shows 56% already do not.

C Patch Manager

Patches the OS and does nothing about the hardening, agents and configuration the golden image carries.

B Delete other AMIs

Breaks running Auto Scaling groups whose launch templates reference them, turning a compliance gap into an outage.

D SCP to prevent, Config rule to detect

Producing a standard and enforcing it are different projects. The SCP stops new non-compliant launches; the Config rule reports what is already running.

THE SIGNAL

The mention of existing SCPs is a hint that the preventive half is available to you. Detect plus prevent, again. Module 27.

An organisation has 180 member accounts. Each has root credentials with a password in a shared vault and an MFA device in a safe. An audit could not locate the MFA devices for 15 accounts, and nobody knew the password for 6. The organisation has a dedicated security account.

The company wants to stop managing root credentials entirely.

What should the solutions architect recommend?

- ☐ A Enable centralized root access management, delete member account root credentials, and register the security account as delegated administrator
- ☐ B Reset all 180 root passwords and order replacement MFA devices
- ☐ C Apply an SCP denying all actions by the root user across the organisation
- ☐ D Record all root credentials in a central register reviewed quarterly

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Centralize and delete

Removes 180 sets of credentials rather than managing them. Accounts created afterwards have none by default, and privileged root actions run centrally.

B Reset and re-order

Restores the problem to working order. still 180 passwords, still 180 devices, and the same audit finding next year.

C SCP denying root

SCPs do not apply to the management account, and the credentials still exist to be stolen or misused.

D Central register

Improves record-keeping of credentials that should not exist. Documentation is not a control.

THE SIGNAL

"Stop managing root credentials entirely" is the phrase. The best security improvement is usually deletion, not better custody. Module 28.

A review finds 74 IAM roles carrying AdministratorAccess, granted between 2019 and 2023. The owning teams refuse removal because they cannot prove nothing would break, and the security team cannot prove the permissions are unused. The company needs to reduce this without causing outages.

CloudTrail has been enabled organisation-wide throughout.

What should the solutions architect recommend?

- ☐ A Remove AdministratorAccess from all 74 roles and restore it where things break
- ☐ B Apply a permissions boundary to all 74 roles and consider the finding closed
- ☐ C Use Access Advisor and IAM Access Analyzer unused access findings to establish what is actually used, then generate least-privilege policies from CloudTrail
- ☐ D Leave them in place and monitor with GuardDuty

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Remove and see what breaks
Discovers dependencies through outages, and it is exactly why the teams are refusing, so it will not be agreed.
- B

 Boundary, finding closed
A useful ceiling that does not reduce what the roles are granted. The audit finding stands.
- C

 Measure use, then generate policies
Replaces an argument about intent with evidence about use. CloudTrail has been on throughout, so policy generation has the history it needs.
- D

 Monitor with GuardDuty
Detects misuse after it happens. Seventy-four administrator roles remain, which was the finding.

THE SIGNAL

The teams' objection is reasonable, and evidence answers it. The CloudTrail detail is what makes policy generation possible. Module 28.

A security review of a GuardDuty deployment enabled organisation-wide in 2022 finds no malware findings for S3 uploads and no runtime findings for EC2 or containers, despite both capabilities appearing in current AWS documentation. GuardDuty reports as enabled and healthy in all 90 accounts.

No suppression rules exist.

What is the MOST likely explanation?

- ☐ A GuardDuty must be disabled and re-enabled to pick up new capabilities
- ☐ B Protection plans released after GuardDuty was first enabled are not turned on automatically for existing customers
- ☐ C Those capabilities require Security Hub to be enabled first
- ☐ D The findings exist but fall below the default severity filter

◆ PAUSE AND DECIDE

What is the MOST likely explanation?

- A

Disable and re-enable
Would lose finding history and is unnecessary. Plans can be enabled individually at any time.
- B

New plans are not auto-enabled
Documented behaviour. On first enabling, all plans switch on except Runtime Monitoring; anything launched later stays off, while the console still reports healthy.
- C

Requires Security Hub
Security Hub aggregates GuardDuty findings. It is not a prerequisite for generating them.
- D

Severity filtering
Would hide low-severity findings, not whole categories. Malware findings are not low severity.

THE SIGNAL

Enabled and healthy means it is running, not that it is watching everything it could. A 2022 deployment lags by years. Module 29.

GuardDuty reports an EC2 instance communicating with a known command-and-control server. The runbook says terminate immediately. Legal requires evidence preserved for possible prosecution, and the response team needs to know what the attacker accessed. The instance has an instance profile with access to three S3 buckets.

The instance is in an Auto Scaling group.

What should the response be?

- ☐ A Terminate the instance immediately, as the runbook specifies
- ☐ B Stop the instance and hand it to the forensics team
- ☐ C Remove the instance profile and leave the instance running
- ☐ D Detach it from the Auto Scaling group, apply an isolation security group, snapshot the volumes, revoke the role's active sessions, and copy evidence to a forensics account

◆ PAUSE AND DECIDE

What should the response be?

- A

 Terminate
Stops the attack and destroys volumes, memory and every trace. Legal's requirement is not met.
- B

 Stop the instance
The near miss. Volumes survive, and memory is lost the moment it stops, which is where much of the useful evidence lives.
- C

 Remove the profile
Leaves the instance talking to the attacker, and credentials already stolen keep working until they expire.
- D

 Detach, isolate, snapshot, revoke, preserve
Detaching stops the ASG replacing it mid-investigation. Isolation stops the traffic, and revoking sessions handles credentials already taken off the box.

THE SIGNAL

Two details: revoking sessions is what people forget, and detaching from the ASG stops it terminating your evidence. Module 29.

An Auto Scaling group spans three Availability Zones and is sized so that all three together exactly meet peak demand. The recovery plan for an AZ failure is that Auto Scaling launches replacements in the surviving zones. During a recent AZ event those launches were throttled and the service was degraded for 40 minutes.

The business will fund additional capacity.

What should the solutions architect recommend?

- ☐ A Increase the Auto Scaling group's maximum size
- ☐ B Add a fourth Availability Zone to the group
- ☐ C Pre-provision capacity so that any two zones can serve full demand, making the design statically stable
- ☐ D Reduce the health check grace period so replacements come into service sooner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Raise the maximum

The group was not at its ceiling. It was throttled, which is a different failure, and a higher maximum does not obtain capacity.

C Pre-provision for N-1

Nothing needs creating when a zone fails, so the control plane is off the recovery path entirely. That is static stability.

B A fourth AZ

Reduces the share lost to one zone, and still requires launching instances during the event. The dependency remains.

D Shorter grace period

Speeds up instances that did launch. The problem was launches being throttled, not being slow to become healthy.

THE SIGNAL

"Throttled" is the giveaway. Every customer's recovery plan fired at once and yours was queuing behind them. Module 30.

An Aurora PostgreSQL cluster is reported as slow. Writer CPU utilisation sits at 18%. Database load is high, and the breakdown shows most sessions waiting on lock events rather than CPU or I/O. One UPDATE statement dominates the load.

The team has proposed a larger instance class and additional read replicas.

What should the solutions architect recommend?

- ☐ A Scale to a larger instance class as proposed
- ☐ B Investigate and fix the contended UPDATE, since sessions are waiting on locks rather than resources
- ☐ C Add read replicas to spread the load
- ☐ D Increase the cluster's provisioned IOPS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Larger instance
More CPU for a database using 18% of what it has. A session waiting on a lock waits exactly as long on a bigger machine.
- C

 Read replicas
Move read traffic. The contention is on an UPDATE, which only ever runs on the writer.
- B

 Fix the contended UPDATE
Lock waits are contention, not capacity. The wait-event breakdown has already named the statement, so the fix is in the query or the transaction.
- D

 More IOPS
The wait events are locks, not I/O. Faster storage does not shorten a wait for another transaction to commit.

THE SIGNAL

Lock contention is the one bottleneck no hardware fixes. That is why wait events matter more than CPU. Module 31.

A company bought a large three-year Compute Savings Plan in February, sized to its On-Demand spend at the time. In April it completed a rightsizing programme that cut compute consumption by 28%. Savings Plans utilisation has sat at 73% since, and finance is asking why savings are below forecast.

The two projects were run by different teams.

What is the situation, and what should change?

- ☐ A Over-committed: the plan was sized to pre-rightsizing usage. Size future commitments to the baseline that will remain after planned changes
- ☐ B Under-committed: buy an additional Savings Plan to raise coverage
- ☐ C Reverse the rightsizing so utilisation recovers
- ☐ D Convert the Compute Savings Plan to an EC2 Instance Savings Plan

◆ PAUSE AND DECIDE

What is the situation, and what should change?

- A

 Over-committed; size to the surviving baseline
Utilisation of 73% means 27% of a three-year commitment buys nothing. Two teams optimised independently and the second undercut the first.
- B

 Buy more
Buying more commitment when a quarter of the existing one is unused makes it worse. Low utilisation means too much, not too little.
- C

 Reverse the rightsizing
Runs oversized instances for three years to justify a commitment. It optimises the metric rather than the bill.
- D

 Convert the plan type
Savings Plans cannot be converted between types. Convertible Reserved Instances can be exchanged; Savings Plans cannot.

THE SIGNAL

Two cost programmes run independently subtracted from each other. That coordination failure is the real lesson. Module 32.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2	Module 23 – Observability foundations
Q3, Q4	Module 24 – Tracing, canaries and RUM
Q5	Module 25 – Automated remediation
Q6, Q7	Module 26 – Systems Manager operations
Q8, Q9, Q10	Modules 27–28 – Deployment and security posture
Q11, Q12, Q13, Q14, Q15	Modules 29–32 – Threats, reliability, performance, cost

ON THE EXAM

Questions 6, 9 and 11 test capabilities newer than most study material. Missing those is a currency gap, not a knowledge gap.